

Do Brain-Inspired Mechanisms Help Continual Reinforcement Learning?

A controlled ablation study with spiking agents

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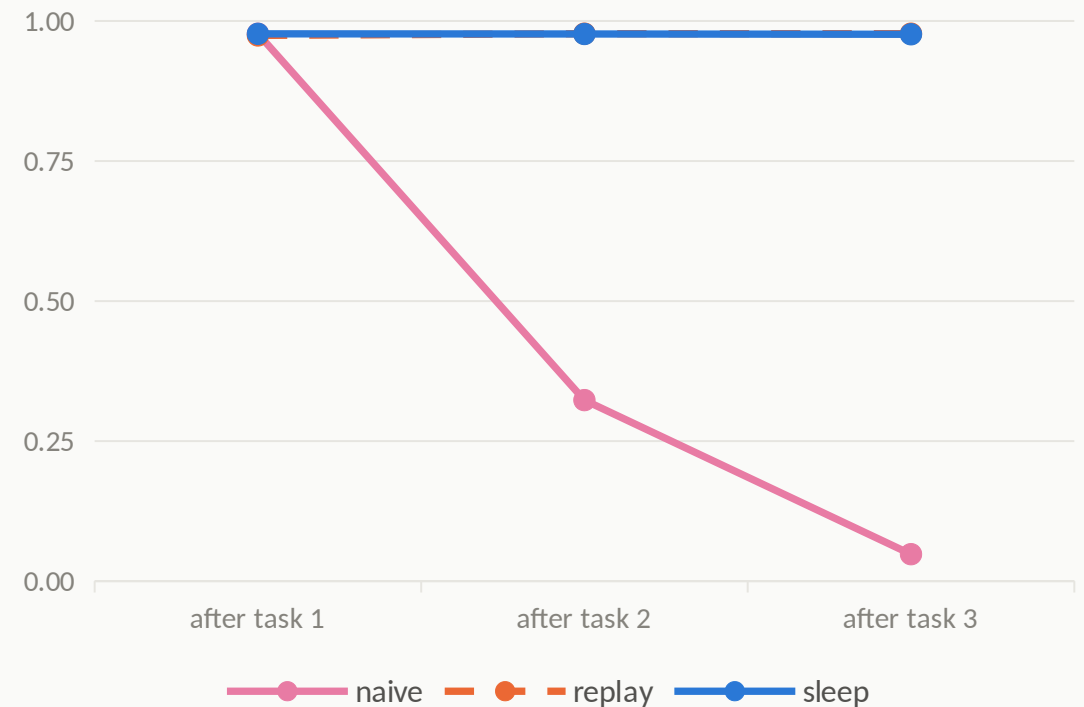
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Learning a new task erases the old one

- An agent learns tasks one after another: fetch the red ball, then the green key, then the blue box
- Plain fine-tuning overwrites what earlier tasks needed: catastrophic forgetting
- Standard fixes: penalise weight changes (EWC), replay stored experience, or one network per task
- Brain-inspired fixes (spikes, local plasticity, sleep) are popular, but are usually added all at once

Return on task 1 as later tasks are learned

CNN agent, 3 tasks, mean of 3 seeds; replay (dashed) and sleep overlap near 1.0



Which brain-inspired mechanisms actually help?

Switch each mechanism on alone and compare it with a control that is not brain-inspired but matches its resources.

Spiking encoder

vs same-shape CNN, matched operations

STDP (local plasticity)

vs a random update of the same size

Homeostasis

vs the same agent without it

DFA: learning without backprop

vs ordinary backpropagation

Sleep-like consolidation

vs replay with an exactly matched replay budget

One shared network

vs one network per task, at equal total memory

Transformer working memory

vs no memory, and a stack of recent frames

Negative results count: the aim is to find out which mechanisms contribute, not to make all of them win.

Related work

● Continual RL

EWC: protect important weights (Kirkpatrick et al., 2017)

CLEAR: replay + behavioural cloning, a very strong baseline (Rolnick et al., 2019)

PackNet: isolate parameters per task (Mallya & Lazebnik, 2018)

● Sleep and consolidation

Sleep-like unsupervised replay recovers forgotten tasks in classifiers (Tadros et al., 2022)

Ours: offline replay + distillation in RL, compared with CLEAR-style replay

● Spiking networks for RL

Surrogate gradients make SNNs trainable (Neftci et al., 2019)

The surrogate slope matters more in RL (Van den Berghe et al., 2025)

Spiking decision transformers (Pandey & Biswas, 2025)

● Local learning rules

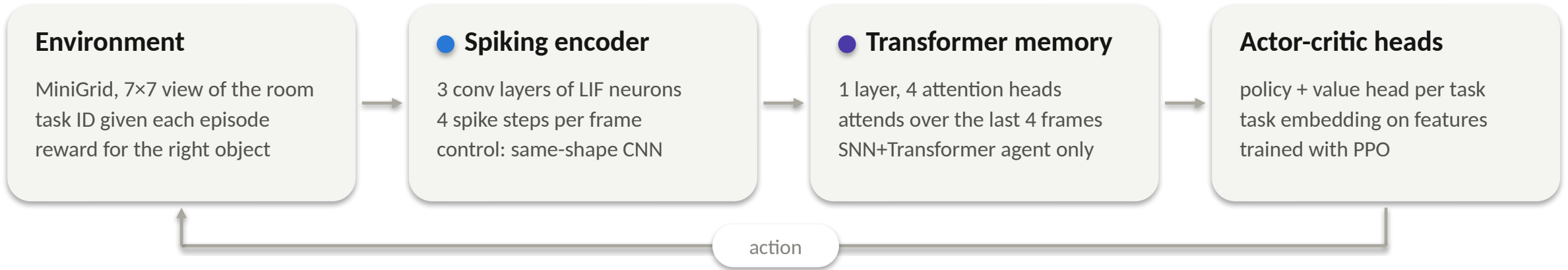
Three-factor, reward-modulated STDP (Frémaux & Gerstner, 2016)

Direct feedback alignment: no weight transport (Nøkland, 2016)

Homeostatic plasticity (Turrigiano & Nelson, 2004)

Gap: these mechanisms are usually tested together, against an unmodified baseline, so it is unclear which one did the work.

System architecture



Continual training of the sleep variant: wake and sleep phases alternate on the same shared network



Experimental setup and fairness controls

● Setup

- MiniGrid: fetch3 / fetch5 sequences (task ID given), DoorKey-6x6, Memory S7/S11/S13
- PPO, small agents (159,864–327,800 parameters)
- Metrics: final accuracy (ACC), forgetting, learning speed (AUC), solve rate

3–10

seeds per arm

289

PPO runs

110

single-thread CPU hours,
no GPU

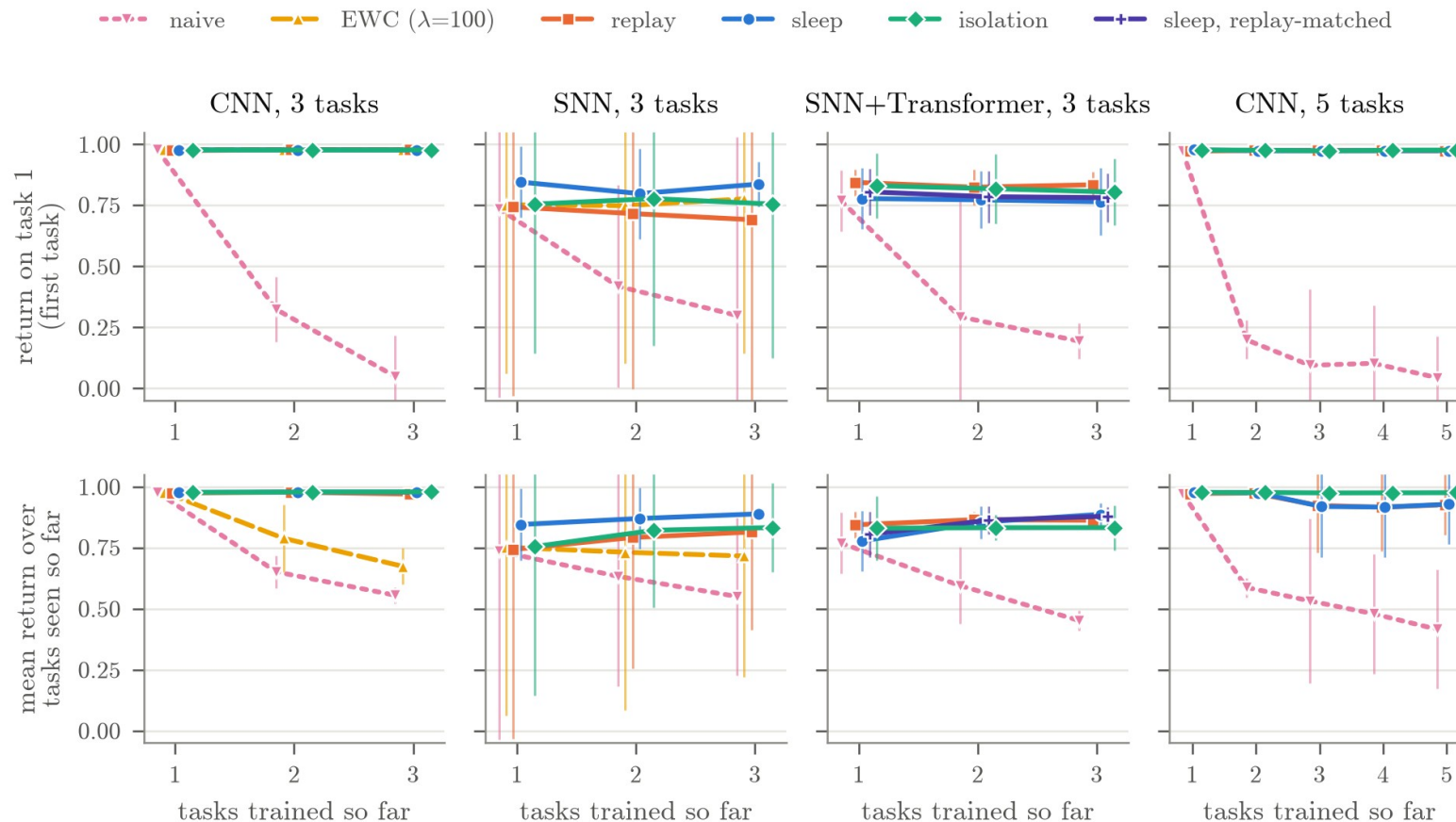
● Fairness controls

- Same frames, seeds and learning rate in every arm of a comparison
- STDP vs a random update of exactly the same size
- Sleep vs replay with the number of replayed samples matched exactly
- Memory counted as parameters plus replay buffer, not parameters alone
- Pre-registered tests; Welch / Fisher; Holm correction; 95% CIs

Results at a glance: mostly no, with two exceptions

MECHANISM	VERDICT	KEY NUMBER (95% CI)
Spiking encoder	⚡ Fewer operations, not more accuracy	works at 16.6k ops/frame where the CNN needs 33k; 5.2× CPU cost
Local learning rule (STDP)	○ No effect	vs random update: $\Delta\text{AUC} +0.033$ $[-0.260, +0.326]$, $p = 0.818$
Homeostasis, without backprop	🕒 Faster learning, not more reliable	$\Delta\text{AUC} +0.326$ $[+0.101, +0.551]$, Holm $p = 0.017$; solved 7/10 vs 2/10 (n.s.)
Homeostasis, with backprop	○ Not significant after correction	$\Delta\text{AUC} +0.200$ $[+0.024, +0.377]$; Holm $p = 0.117$
Sleep-like consolidation	○ Ties replay	sleep – replay $\Delta\text{ACC} +0.017$ $[-0.026, +0.060]$, $p = 0.415$
One shared network	🕒 Only with small replay buffers	200 states/task: ties isolation in 0.89 vs 1.92 MB
Transformer working memory	? Untested: no usable benchmark	no learnable memory benchmark within budget

Sleep and replay stop forgetting; sleep does not beat replay



0.555

naive fine-tuning, final ACC (CNN, 3 tasks)

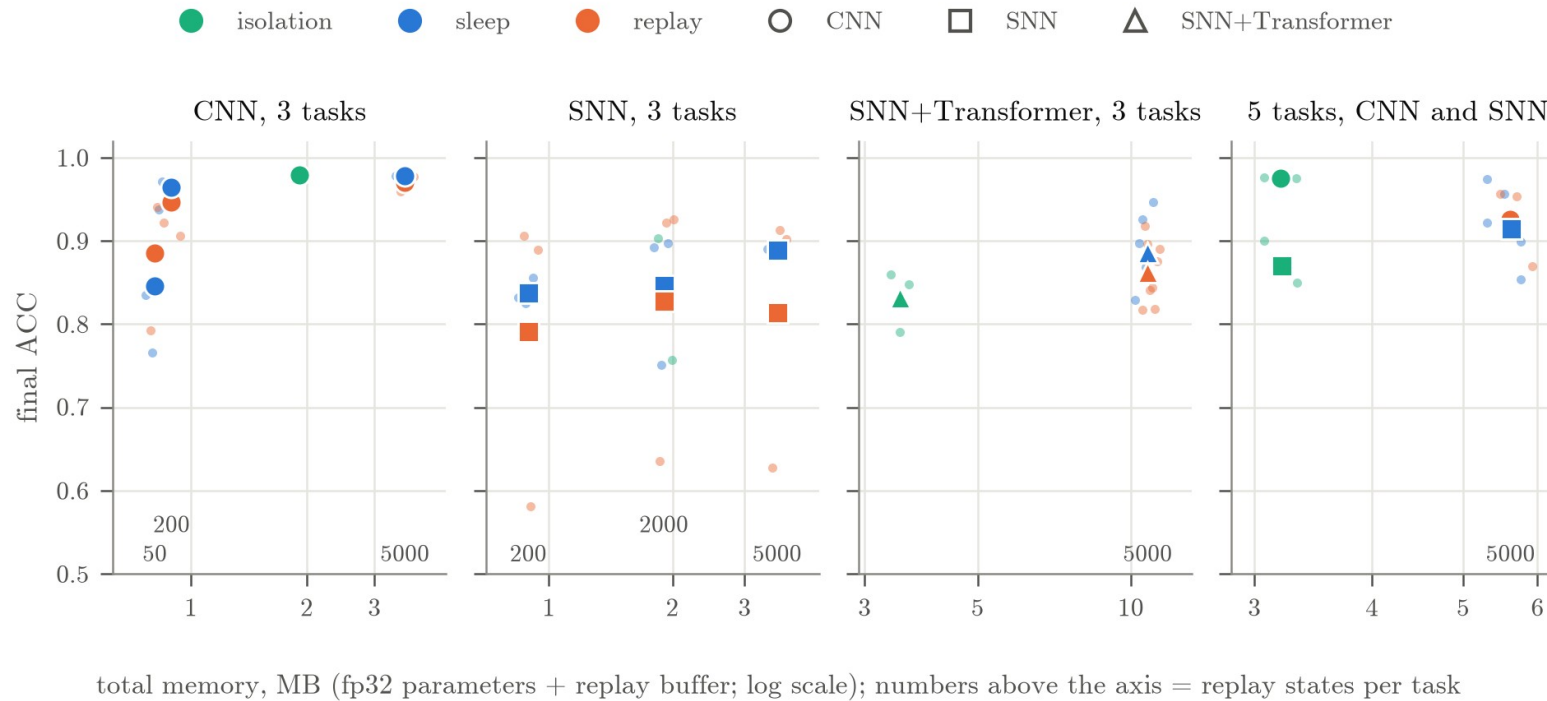
0.970 · 0.978

replay · sleep, final ACC

+0.017

sleep – replay at an exactly matched replay budget,
95% CI $[-0.026, +0.060]$, $p = 0.415$

A shared network wins only with a small replay buffer



Total memory = fp32 parameters + replay buffer. Numbers above the axis: replay states stored per task.

46%

of the memory: SNN sleep with 200 states/task ties isolation (0.838 vs 0.833; 0.89 vs 1.92 MB)

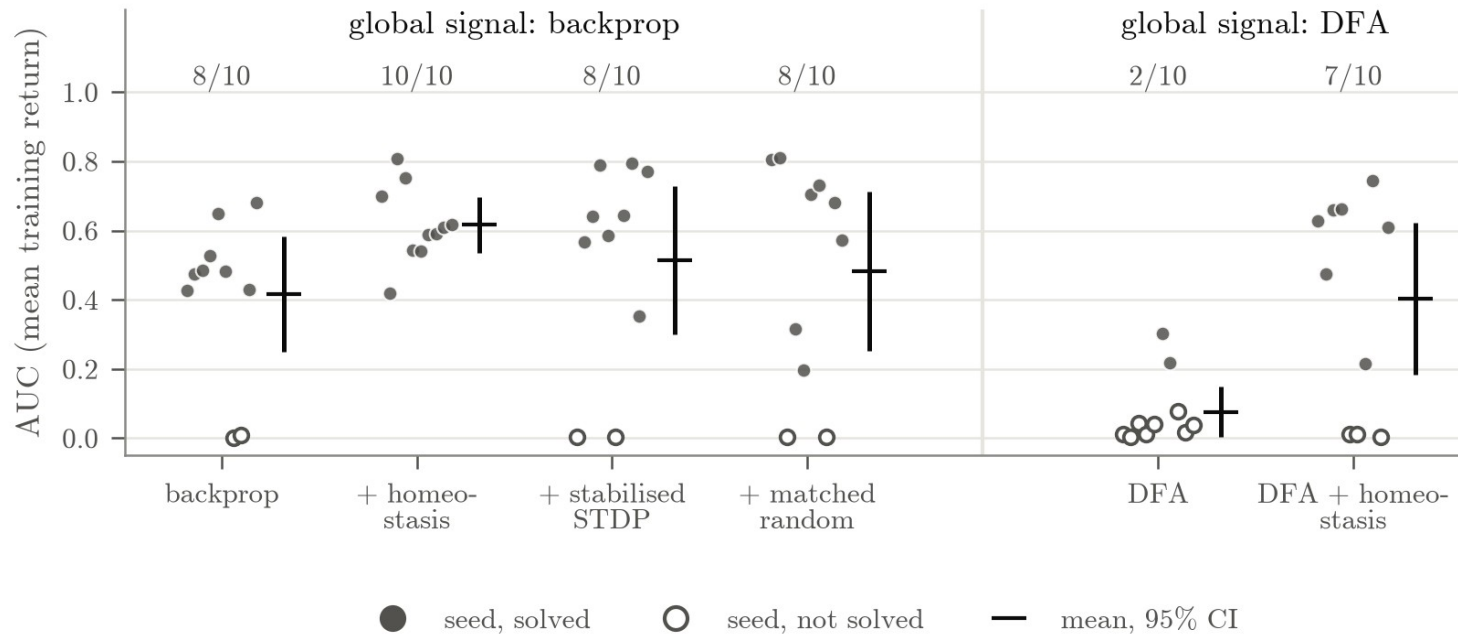
2.5–3.3×

more accuracy per parameter for the shared network

1.8–2.9×

more accuracy per MB for isolation at the default 5,000 states/task

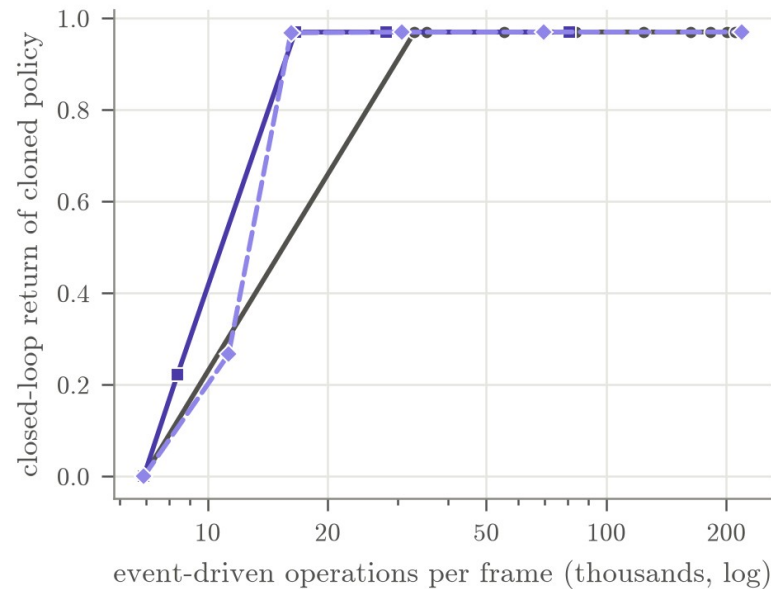
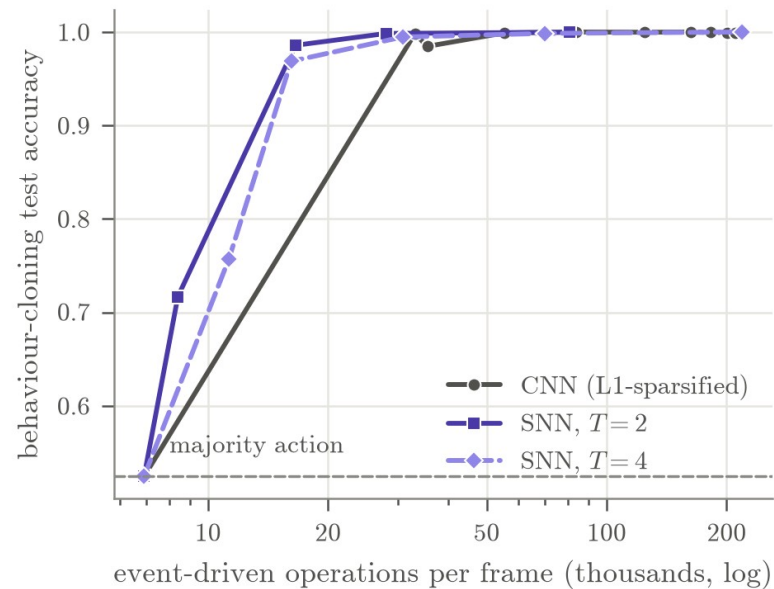
STDP adds nothing; homeostasis speeds up DFA



- **STDP $\approx$ random update:** $\Delta\text{AUC} +0.033$ $[-0.260, +0.326]$, $p = 0.818$
- **Unstabilised STDP hurts:** solved 0/3 vs 3/3 with backprop (runaway firing)
- **Homeostasis + backprop:** raw $p = 0.029$, Holm $p = 0.117$: not significant
- **Homeostasis + DFA:** $\Delta\text{AUC} +0.326$ $[+0.101, +0.551]$, Holm $p = 0.017$; solved 7/10 vs 2/10 (Holm $p = 0.070$)

DoorKey-6x6, 10 seeds per arm, pre-registered. Dots: seeds (filled = solved); bars: mean and 95% CI; numbers: seeds solved.

Spiking encoder: fewer operations, not more accuracy



16.6k

ops/frame: the SNN still works (accuracy 0.986)

33k

ops/frame: below this the sparsified CNN stops working

5.2×

CPU time per training step (80 vs 16 ms); energy savings are only an estimate

Behaviour cloning of a DoorKey-6x6 teacher, activity penalty swept; 2 seeds per point. Operations = synaptic events (SNN) or non-zero multiply-accumulates (sparsified CNN).

What didn't survive controls, and why

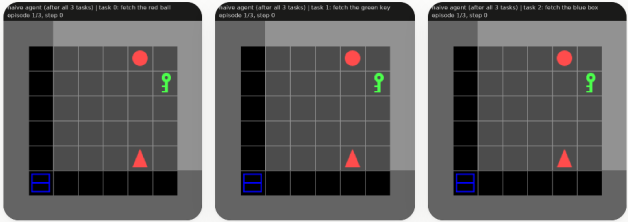
EARLIER FINDING	EVIDENCE THEN	AFTER THE CONTROL	CAUSE
Sleep beats replay (SNN+Transformer)	p = 0.072 with 3 seeds	tie at a matched replay budget: p = 0.415, 8 seeds	seeds; replay budget
Shared weights far more efficient	2.5–3.3× per parameter	isolation 1.8–2.9× better per MB	memory accounting
Sleep far beats isolation, 5 tasks (SNN)	gap +0.303 [+0.148, +0.457]	gap +0.044 [−0.009, +0.098] at a fair budget	training budget
Homeostasis helps backprop	p = 0.088 (3 vs 5 seeds)	Holm p = 0.117 with 10 seeds	seeds; multiple tests
Homeostasis makes DFA work	solved 3/3 vs 1/5	7/10 vs 2/10, p = 0.070 (AUC effect survives)	seeds
Transformer helps on a memory task	S7 pilot 0.970 vs 0.465	no memory needed on S7 (memoryless CNN 0.915)	benchmark validity

Each apparent win disappeared once the resource it quietly used more of was matched.

Live demo

Naive fine-tuning

seed 1 · final accuracy 0.541 · mean of 3 seeds 0.555



Task 1: red ball
✓ X X (1 of 3)

Task 2: green key
✓ ✓ X (2 of 3)

Task 3: blue box
✓ ✓ ✓ (3 of 3)

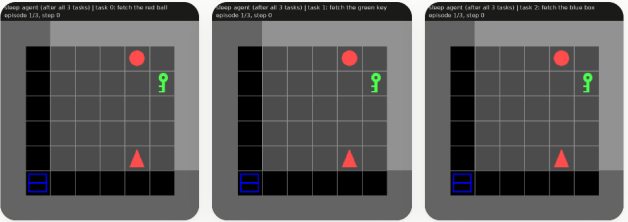
Task 1: the first clip happens to succeed (1 of 3 episodes here).
Over 100 evaluation episodes this agent's task-1 return is 0.126;
the matrix gives the real average.

	task 1	task 2	task 3
after task 1	0.977	0.275	0.119
after task 2	0.265	0.979	0.141
after task 3	0.126	0.542	0.956

Rows: after training task i. Columns: mean return on task j. The
bottom row is final performance; its mean is the final accuracy.

Sleep-like consolidation

seed 1 · final accuracy 0.978 · mean of 3 seeds 0.978



Task 1: red ball
✓ ✓ ✓ (3 of 3)

Task 2: green key
✓ ✓ ✓ (3 of 3)

Task 3: blue box
✓ ✓ ✓ (3 of 3)

9 of 9 episodes shown succeed, in line with the matrix: over 100
evaluation episodes per task, final returns are 0.978, 0.979,
0.979.

	task 1	task 2	task 3
after task 1	0.978	0.227	0.165
after task 2	0.979	0.980	0.080
after task 3	0.978	0.979	0.979

Rows: after training task i. Columns: mean return on task j. The
bottom row is final performance; its mean is the final accuracy.

1. Results dashboard

open demo/index.html (one file, works offline)
scorecard, agents, forgetting and memory charts

2. Run an agent on CPU

```
python demo/live_episode.py  
--agent naive --task 0 --window  
then the same with --agent sleep  
prints each step and the real accuracy matrix
```

Limitations and future work

● Limitations

- Task ID is given (task-incremental), so this says nothing about task-agnostic learning
- Small scale: MiniGrid, $\leq 327,800$ parameters, 3–5 tasks
- Few seeds: many continual arms have 3, so CIs are wide
- Energy is an estimate from operation counts; no hardware was measured
- Transformer memory untested: no benchmark was learnable in budget

● Future work

- A memory benchmark agents can learn (the cue seen on the way to the choice)
- **Compressed replay (idea, untested):** the buffer size decides the shared-vs-isolation trade-off, so shrink it: quantised targets, fewer but better-chosen states, or generative replay
- Homeostasis target sweep: is the DFA effect robust?
- More seeds for the operation-count frontier

Mostly no, with two specific exceptions.

1

No mechanism beat its matched control

on final accuracy or forgetting: STDP $\approx$ random update, sleep $\approx$ replay

2

Two exceptions

homeostasis speeds up learning without backprop; a shared network is more memory-efficient with a small replay buffer

3

The lesson is methodological

every apparent win vanished once the resource it quietly used more of (seeds, frames, memory, replayed samples) was matched

Thank you. Questions?

Backup: final accuracy and forgetting by method

	CNN, 3 tasks	SNN, 3 tasks	SNN+Transformer, 3 tasks	CNN, 5 tasks
Final ACC (mean $\pm$ SD)				
naive	0.555 $\pm$ 0.013	0.550 $\pm$ 0.130	0.452 $\pm$ 0.016	0.416 $\pm$ 0.098
EWC ($\lambda=100$)	0.674 $\pm$ 0.030	0.716 $\pm$ 0.199	–	–
replay	0.970 $\pm$ 0.009	0.814 $\pm$ 0.162	0.862 $\pm$ 0.038	0.926 $\pm$ 0.049
sleep	0.978 $\pm$ 0.001	0.889 $\pm$ 0.003	0.887 $\pm$ 0.045	0.928 $\pm$ 0.065
sleep, matched	–	–	0.879 $\pm$ 0.043	–
isolation	0.979 $\pm$ 0.001	0.833 $\pm$ 0.073	0.832 $\pm$ 0.037	0.976 $\pm$ 0.000
FORGET (mean)				
naive	0.631	0.321	0.536	0.612
replay	0.002	0.023	0.019	0.006
sleep	0.001	0.006	0.006	0.003
sleep, matched	–	–	0.020	–

Seeds: 3 per arm, except SNN+Transformer (replay 8, sleep 6, sleep matched 8). Paper Table 3.

Backup: pre-registered local-plasticity contrasts

Contrast	Δ solve rate [95% CI]	Fisher p (Holm)	Δ AUC [95% CI]	Welch p (Holm)
STDP vs homeostasis alone	-0.20 [-0.51, +0.11]	0.474 (1.000)	-0.101 [-0.323, +0.121]	0.338 (0.724)
STDP vs matched random	+0.00 [-0.34, +0.34]	1.000 (1.000)	+0.033 [-0.260, +0.326]	0.818 (0.818)
homeostasis vs backprop	+0.20 [-0.11, +0.51]	0.474 (1.000)	+0.200 [+0.024, +0.377]	0.029 (0.117)
random vs homeostasis	-0.20 [-0.51, +0.11]	0.474 (1.000)	-0.134 [-0.372, +0.104]	0.241 (0.724)
DFA + homeostasis vs DFA	+0.50 [+0.07, +0.74]	0.070 (0.070)	+0.326 [+0.101, +0.551]	0.009 (0.017)

DoorKey-6x6, 10 seeds per arm. First four rows: one Holm family per statistic (backprop as the global signal). Last row: the DFA family (pre-registered separately). Newcombe CIs for solve rates, Welch for AUC.

Backup: protocol constants

PPO	16 envs $\times$ 128 steps, 4 epochs, minibatch 256, clip 0.2, $\gamma = 0.99$, GAE $\lambda = 0.95$, KL stop 0.03
Learning rate	0.001 (CNN, SNN); 0.0003 (SNN+Transformer)
Frames per task	150k (CNN, SNN); 400k (SNN+Transformer); 450k (5-task SNN, fair budget)
SNN	4 time steps, learnable leak (init 0.9), surrogate slope 25, membrane reset each frame
Plasticity	$\alpha = 0.03$; homeostasis target 0.15, rate 0.01
Sleep	every 25 updates + task end; 700 steps per phase; 5,000 states per task
Evaluation	100 episodes per task (continual); 200 (single task)
Memory accounting	188 B/state (CNN, SNN); 632 B/state (SNN+Transformer)

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